Liang Yu

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German and South African dual citizen

Education

Doctor of Philosophy | ***Physics*** — Jun 2019
Massachusetts Institute of Technology (MIT) — Cambridge, MA, USA

- Thesis: Elucidating the Evolution Histories of Giant Planets with *K2* and *TESS*
- Research areas: Exoplanet detection and characterization
- GPA: 5.00 / 5.00

Bachelor of Science | ***Astronomy & Physics*** — May 2014
Yale University — New Haven, CT, USA

- Graduated summa cum laude, with exceptional distinction in the Astronomy & Physics major
- GPA: 4.00 / 4.00

Relevant Experience

Lead AI Architect — Nov 2024 – present
BMW Group — Munich, Germany

- Lead the development of onboard machine learning platform to support custom inference frameworks, scheduling, agentic workflows and monitoring on constrained in-vehicle hardware.
- Evaluate and benchmark emerging generative AI models and accelerators, with a focus on deploying small language models (SLMs) on NPUs for production-grade automotive use cases.
- Fine tune and optimize SLMs for heterogeneous hardware and frameworks for in-vehicle voice assistant use cases. Ensure a balance between efficient execution and reliability.
- Lead hiring and organizational development for the ML Core Platform team in India, including technical mentorship and performance management.
- Advise executive leadership and board stakeholders on AI strategy, influencing investment decisions and product direction across global markets, with deep focus on China (technology trends, competitors, and regulatory landscape).

Data Science Team Lead, Big Loop and Advanced Systems — Oct 2020 – Oct 2024
Volkswagen Group CARIAD SE (Audi AG until Apr 2021) — Ingolstadt, Germany

- Hired and supervised a group of 15+ software engineers and data scientists. Established engineering best practices, development workflows, and a production-grade codebase for data-centric AI systems for autonomous driving.
- Led the development of in-vehicle data selection triggers for camera images, enabling efficient acquisition of high-value training data by detecting high-uncertainty and novel objects in real-world driving scenarios.
- Defined product vision and roadmap for data selection triggers, communicated with cross-functional stakeholders and oversaw successful deployment of triggers across the VW Group fleet.
- Contributed as data scientist and developer to the research, testing and optimization of uncertainty quantification and anomaly detection algorithms to solve the "long-tail" problem, as well as active learning experiments for perception tasks (semantic segmentation, object detection).
- Influenced executive decision-making by developing statistical models to estimate the required empirical test effort for autonomous vehicles.

Data Scientist, Upstream Integrated Solutions — Aug 2019 – Sep 2020
ExxonMobil — Spring, TX, USA

- Developed smoothing filter-based models for the calibration and interpretation of time series data from acoustic sand monitors and seismic sensors, and a web API for visualizing sensor data and sending real-time alerts to users.
- Automated predictions of drilling outcomes of basins using supervised learning. Matched the accuracy of predictions by experts in geophysics.
- Researched and applied polarization filtering and spectral analysis to detect low signal-to-noise microseismic events in time series data from three-component seismic sensors.

Graduate Research Student, Exoplanet Group Sep 2014 – Jun 2019
MIT, Department of Physics Cambridge, MA, USA

- Co-authored Astronet-Triage and Astronet-Vetting, convolutional neural networks that identify planet candidates and false positives in time series data from large-scale surveys. Currently used by the Transiting Exoplanets Survey Satellite (TESS) mission.
- Constructed pipeline to extract stellar flux, remove systematic noise with Gaussian process regression and identify planet-like signals from raw data taken by the *K2* space mission. Discovered 81 confirmed planets and candidates.

Intern, Big Data Strategies Jun 2016 – Aug 2016
Totsa Total Oil Trading SA Geneva, Switzerland

- Researched supervised learning techniques to improve ship tracking accuracy, predict ship destinations based on partial trajectories, and identify patterns and anomalies in global ship movement.
- Developed algorithms to extract the most common ship trajectories between ports and calculate the length and duration of a typical trip.
- Simplified ship trajectories and removed outliers with clustering methods for easier processing.

Student Supervision

ElHayani, M. Master's thesis — Technical University of Munich 2023-2024
Title: Real-Time Object Detection Uncertainty Quantification Using Augmented Images for Autonomous Vehicles

Technical Skills

Data science/ML: Edge AI, computer vision, generative AI (LLM/VLM), agentic AI, RAG, statistics, Bayesian inference, uncertainty quantification, time series analysis
Programming: Experienced with Python (incl. PyTorch, Keras, TensorFlow, Scikit-learn, LangChain, LangGraph) and SQL; basic knowledge of Bash, R, MATLAB and Mathematica
Technologies: Git, Azure
Languages: fluent English and Mandarin Chinese; intermediate German, French, Spanish, and Italian

Invited Talks

Finding the Unknown Unknowns: Intelligent Data Collection for Autonomous Driving Development 2022
WeAreDevelopers World Congress

An Uncertainty-Based Corner Case Detection Framework for Autonomous Driving 2021
Sensors & IIoT - Manufacturing, Automation and Robotics

Model-Enhanced Machine Learning for Astronomical Discovery 2016
University of Toronto

Tests of the Planetary Hypothesis for PTFO 8-8695b 2015
Boston University

Tests of the Planetary Hypothesis for PTFO 8-8695b 2015
Harvard University

Selected Publications

Yu, L.; Vanderburg, A.; Huang, C. X.; Shallue, C. J.; Crossfield, I. J. M. et al. (2019), *Identifying Exoplanets with Deep Learning III: Automated Triage and Vetting of TESS Candidates*, the Astronomical Journal, 158, 25.

Yu, L.; Zhou, G.; Rodriguez, J. E.; Huang, C. X.; Vanderburg, A. et al. (2018), *EPIC 246851721 b: A Tropical Jupiter Transiting a Rapidly Rotating Star in a Well-Aligned Orbit*, the Astronomical Journal, 156, 250.

Yu, L.; Rodriguez, J. E.; Eastman, J. D.; Crossfield, I. J. M.; Shporer, A. et al. (2018), *Two Warm, Low-density Sub-Jovian Planets Orbiting Bright Stars in K2 Campaigns 13 and 14*, the Astronomical Journal, 156, 127.

Yu, L.; Crossfield, I. J. M.; Schlieder, J. E.; Kosiarek, M. R.; Feinstein, A. D. et al. (2018), *Planetary Candidates from K2 Campaign 16*, the Astronomical Journal, 156, 22.

Yu, L.; Winn, J. N.; Gillon, M.; Albrecht, S.; Rappaport, S. et al. (2015), *Tests of the Planetary Hypothesis for PTFO 8-8695b*, the Astrophysical Journal, 812, 48.

Yu L.; Nelson, K.; Nagai, D. (2015), *The Influence of Mergers on Scatter and Evolution in Sunyaev-Zel'dovich Effect Scaling Relations*, the Astrophysical Journal, 807, 12.